



MATH240: Laplace Transforms

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Laplace transform of exponentials

1. Find the Laplace transform of $f(t) = 6e^{-4t}$.

$$\mathcal{L}\{6e^{-4t}\}$$

Answer: _____

2. Find the Laplace transform of $f(t) = 1e^{1t}$.

$$\mathcal{L}\{1e^{1t}\}$$

Answer: _____

3. Find the Laplace transform of $f(t) = 4e^{-1t}$.

$$\mathcal{L}\{4e^{-1t}\}$$

Answer: _____

4. Find the Laplace transform of $f(t) = 2e^{2t}$.

$$\mathcal{L}\{2e^{2t}\}$$

Answer: _____

5. Find the Laplace transform of $f(t) = 5e^{-2t}$.

$$\mathcal{L}\{5e^{-2t}\}$$

Answer: _____

6. Find the Laplace transform of $f(t) = 5e^{-3t}$.

$$\mathcal{L}\{5e^{-3t}\}$$

Answer: _____



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7. Find the Laplace transform of $f(t) = 1 \cdot e^{-1 \cdot t}$.

$$\mathcal{L}\{1e^{-1t}\}$$

Answer: _____

8. Find the Laplace transform of $f(t) = 1 \cdot e^{-1 \cdot t}$.

$$\mathcal{L}\{1e^{-1t}\}$$

Answer: _____

Laplace transform of power functions

9. Find the Laplace transform of $f(t) = 1 \cdot t^2$.

$$\mathcal{L}\{1t^2\}$$

Answer: _____

10. A forcing function is $f(t) = 4 \cdot t^3$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{4t^3\}$$

Answer: _____

11. Find the Laplace transform of $f(t) = 3 \cdot t^0$.

$$\mathcal{L}\{3t^0\}$$

Answer: _____

12. A forcing function is $f(t) = 3 \cdot t^1$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{3t^1\}$$

Answer: _____

13. Find the Laplace transform of $f(t) = 2 \cdot t^1$.

$$\mathcal{L}\{2t^1\}$$

Answer: _____

14. A forcing function is $f(t) = 3 \cdot t^1$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{3t^1\}$$

Answer: _____



15. Find the Laplace transform of $f(t) = 1 \cdot t^1$.

$$\mathcal{L}\{1t^1\}$$

Answer: _____

16. A forcing function is $f(t) = 1 \cdot t^2$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{1t^2\}$$

Answer: _____

17. Find the Laplace transform of $f(t) = 2 \cdot t^0$.

$$\mathcal{L}\{2t^0\}$$

Answer: _____

18. A forcing function is $f(t) = 3 \cdot t^2$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{3t^2\}$$

Answer: _____

19. Find the Laplace transform of $f(t) = 4 \cdot t^3$.

$$\mathcal{L}\{4t^3\}$$

Answer: _____

20. A forcing function is $f(t) = 4 \cdot t^1$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{4t^1\}$$

Answer: _____

21. Find the Laplace transform of $f(t) = 1 \cdot t^0$.

$$\mathcal{L}\{1t^0\}$$

Answer: _____

22. A forcing function is $f(t) = 4 \cdot t^3$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{4t^3\}$$

Answer: _____



23. Find the Laplace transform of $f(t) = 3t^0$.

$$\mathcal{L}\{3t^0\}$$

Answer: _____

Laplace transform of sine

24. Find the Laplace transform of $f(t) = 3\sin(4t)$.

$$\mathcal{L}\{3\sin(4t)\}$$

Answer: _____

25. Find the Laplace transform of $f(t) = 1\sin(3t)$.

$$\mathcal{L}\{1\sin(3t)\}$$

Answer: _____

26. Find the Laplace transform of $f(t) = 5\sin(2t)$.

$$\mathcal{L}\{5\sin(2t)\}$$

Answer: _____

27. Find the Laplace transform of $f(t) = 1\sin(3t)$.

$$\mathcal{L}\{1\sin(3t)\}$$

Answer: _____

28. Find the Laplace transform of $f(t) = 4\sin(2t)$.

$$\mathcal{L}\{4\sin(2t)\}$$

Answer: _____

29. Find the Laplace transform of $f(t) = 4\sin(3t)$.

$$\mathcal{L}\{4\sin(3t)\}$$

Answer: _____

30. Find the Laplace transform of $f(t) = 1\sin(1t)$.

$$\mathcal{L}\{1\sin(1t)\}$$

Answer: _____





MATH240: Laplace Transforms

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ANSWER KEY & SOLUTIONS

Topics: Laplace transform of sine, Laplace transform of power functions, Laplace transform of exponentials. All answers verified by independent computation.

Solutions



Laplace transform of exponentials

1. Find the Laplace transform of $f(t) = 6e^{-4t}$.

$$\mathcal{L}\{6e^{-4t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[6e^{-4t}] = 6/(s - -4)$.

→ The denominator shift is $s = -4$.

Answer:
$$= \frac{6}{s - -4}$$

2. Find the Laplace transform of $f(t) = 1e^{1t}$.

$$\mathcal{L}\{1e^{1t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[1e^{1t}] = 1/(s - 1)$.

→ The denominator shift is $s = 1$.

Answer:
$$= \frac{1}{s - 1}$$

3. Find the Laplace transform of $f(t) = 4e^{-1t}$.

$$\mathcal{L}\{4e^{-1t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[4e^{-1t}] = 4/(s - -1)$.

→ The denominator shift is $s = -1$.

Answer:
$$= \frac{4}{s - -1}$$

4. Find the Laplace transform of $f(t) = 2e^{2t}$.

$$\mathcal{L}\{2e^{2t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[2e^{2t}] = 2/(s - 2)$.

→ The denominator shift is $s = 2$.

Answer:
$$= \frac{2}{s - 2}$$

5. Find the Laplace transform of $f(t) = 5e^{-2t}$.

$$\mathcal{L}\{5e^{-2t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[5e^{-2t}] = 5/(s - -2)$.

→ The denominator shift is $s = -2$.

Answer:
$$= \frac{5}{s - -2}$$



6. Find the Laplace transform of $f(t) = 5e^{-3t}$.

$$\mathcal{L}\{5e^{-3t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[5e^{-3t}] = 5/(s - -3)$.

→ The denominator shift is $s = -3$.

Answer:
$$= \frac{5}{s - -3}$$

7. Find the Laplace transform of $f(t) = 1e^{-1t}$.

$$\mathcal{L}\{1e^{-1t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[1e^{-1t}] = 1/(s - -1)$.

→ The denominator shift is $s = -1$.

Answer:
$$= \frac{1}{s - -1}$$

8. Find the Laplace transform of $f(t) = 1e^{-1t}$.

$$\mathcal{L}\{1e^{-1t}\}$$

→ Formula: $\mathcal{L}[e^{bt}] = 1/(s-b)$.

→ $\mathcal{L}[1e^{-1t}] = 1/(s - -1)$.

→ The denominator shift is $s = -1$.

Answer:
$$= \frac{1}{s - -1}$$



Laplace transform of power functions

9. Find the Laplace transform of $f(t) = 1 \cdot t^2$.

$$\mathcal{L}\{1t^2\}$$

→ Formula: $\mathcal{L}\{t^n\} = n! / s^{(n+1)}$.

→ $\mathcal{L}\{1t^2\} = 1 \cdot 2 / s^3 = 2 / s^3$.

→ Numerator coefficient: 2.

Answer: $= \frac{2}{s^3}$

10. A forcing function is $f(t) = 4 \cdot t^3$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{4t^3\}$$

→ $\mathcal{L}\{4 \cdot t^3\} = 4 \cdot 3! / s^4$.

→ Numerator: $4 \cdot 6 = 24$.

→ $F(s) = 24/s^4$.

Answer: $= \frac{24}{s^4}$

11. Find the Laplace transform of $f(t) = 3 \cdot t^0$.

$$\mathcal{L}\{3t^0\}$$

→ Formula: $\mathcal{L}\{t^n\} = n! / s^{(n+1)}$.

→ $\mathcal{L}\{3 \cdot t^0\} = 3 \cdot 1 / s^1 = 3 / s^1$.

→ Numerator coefficient: 3.

Answer: $= \frac{3}{s^1}$

12. A forcing function is $f(t) = 3 \cdot t^1$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{3t^1\}$$

→ $\mathcal{L}\{3 \cdot t^1\} = 3 \cdot 1! / s^2$.

→ Numerator: $3 \cdot 1 = 3$.

→ $F(s) = 3/s^2$.

Answer: $= \frac{3}{s^2}$

13. Find the Laplace transform of $f(t) = 2 \cdot t^1$.

$$\mathcal{L}\{2t^1\}$$

→ Formula: $\mathcal{L}\{t^n\} = n! / s^{(n+1)}$.

→ $\mathcal{L}\{2 \cdot t^1\} = 2 \cdot 1 / s^2 = 2 / s^2$.

→ Numerator coefficient: 2.

Answer: $= \frac{2}{s^2}$



14. A forcing function is $f(t) = 3t^1$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{3t^1\}$$

$$\rightarrow L[3t^1] = 3 \cdot 1! / s^2.$$

$$\rightarrow \text{Numerator: } 3 \cdot 1 = 3.$$

$$\rightarrow F(s) = 3/s^2.$$

Answer: $= \frac{3}{s^2}$

15. Find the Laplace transform of $f(t) = 1t^1$.

$$\mathcal{L}\{1t^1\}$$

$$\rightarrow \text{Formula: } L[t^n] = n! / s^{(n+1)}.$$

$$\rightarrow L[1t^1] = 1 \cdot 1 / s^2 = 1 / s^2.$$

$$\rightarrow \text{Numerator coefficient: } 1.$$

Answer: $= \frac{1}{s^2}$

16. A forcing function is $f(t) = 1t^2$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{1t^2\}$$

$$\rightarrow L[1t^2] = 1 \cdot 2! / s^3.$$

$$\rightarrow \text{Numerator: } 1 \cdot 2 = 2.$$

$$\rightarrow F(s) = 2/s^3.$$

Answer: $= \frac{2}{s^3}$

17. Find the Laplace transform of $f(t) = 2t^0$.

$$\mathcal{L}\{2t^0\}$$

$$\rightarrow \text{Formula: } L[t^n] = n! / s^{(n+1)}.$$

$$\rightarrow L[2t^0] = 2 \cdot 1 / s^1 = 2 / s^1.$$

$$\rightarrow \text{Numerator coefficient: } 2.$$

Answer: $= \frac{2}{s^1}$

18. A forcing function is $f(t) = 3t^2$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{3t^2\}$$

$$\rightarrow L[3t^2] = 3 \cdot 2! / s^3.$$

$$\rightarrow \text{Numerator: } 3 \cdot 2 = 6.$$

$$\rightarrow F(s) = 6/s^3.$$

Answer: $= \frac{6}{s^3}$



19. Find the Laplace transform of $f(t) = 4t^3$.

$$\mathcal{L}\{4t^3\}$$

→ Formula: $L[t^n] = n! / s^{(n+1)}$.

→ $L[4t^3] = 4 \cdot 6 / s^4 = 24 / s^4$.

→ Numerator coefficient: 24.

Answer: $= \frac{24}{s^4}$

20. A forcing function is $f(t) = 4t^1$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{4t^1\}$$

→ $L[4t^1] = 4 \cdot 1! / s^2$.

→ Numerator: $4 \cdot 1 = 4$.

→ $F(s) = 4/s^2$.

Answer: $= \frac{4}{s^2}$

21. Find the Laplace transform of $f(t) = 1t^0$.

$$\mathcal{L}\{1t^0\}$$

→ Formula: $L[t^n] = n! / s^{(n+1)}$.

→ $L[1t^0] = 1 \cdot 1 / s^1 = 1 / s^1$.

→ Numerator coefficient: 1.

Answer: $= \frac{1}{s^1}$

22. A forcing function is $f(t) = 4t^3$. Find the Laplace transform $F(s)$ and identify the numerator coefficient.

$$\mathcal{L}\{4t^3\}$$

→ $L[4t^3] = 4 \cdot 3! / s^4$.

→ Numerator: $4 \cdot 6 = 24$.

→ $F(s) = 24/s^4$.

Answer: $= \frac{24}{s^4}$

23. Find the Laplace transform of $f(t) = 3t^0$.

$$\mathcal{L}\{3t^0\}$$

→ Formula: $L[t^n] = n! / s^{(n+1)}$.

→ $L[3t^0] = 3 \cdot 1 / s^1 = 3 / s^1$.

→ Numerator coefficient: 3.

Answer: $= \frac{3}{s^1}$



Laplace transform of sine

24. Find the Laplace transform of $f(t) = 3\sin(4t)$.

$$\mathcal{L}\{3\sin(4t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

$$\rightarrow L[3\sin(4t)] = 3 \cdot 4/(s^2 + 16) = 12/(s^2 + 16).$$

Answer:
$$= \frac{12}{s^2 + 16}$$

25. Find the Laplace transform of $f(t) = 1\sin(3t)$.

$$\mathcal{L}\{1\sin(3t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

$$\rightarrow L[1\sin(3t)] = 1 \cdot 3/(s^2 + 9) = 3/(s^2 + 9).$$

Answer:
$$= \frac{3}{s^2 + 9}$$

26. Find the Laplace transform of $f(t) = 5\sin(2t)$.

$$\mathcal{L}\{5\sin(2t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

$$\rightarrow L[5\sin(2t)] = 5 \cdot 2/(s^2 + 4) = 10/(s^2 + 4).$$

Answer:
$$= \frac{10}{s^2 + 4}$$

27. Find the Laplace transform of $f(t) = 1\sin(3t)$.

$$\mathcal{L}\{1\sin(3t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

$$\rightarrow L[1\sin(3t)] = 1 \cdot 3/(s^2 + 9) = 3/(s^2 + 9).$$

Answer:
$$= \frac{3}{s^2 + 9}$$

28. Find the Laplace transform of $f(t) = 4\sin(2t)$.

$$\mathcal{L}\{4\sin(2t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

$$\rightarrow L[4\sin(2t)] = 4 \cdot 2/(s^2 + 4) = 8/(s^2 + 4).$$

Answer:
$$= \frac{8}{s^2 + 4}$$

29. Find the Laplace transform of $f(t) = 4\sin(3t)$.

$$\mathcal{L}\{4\sin(3t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

$$\rightarrow L[4\sin(3t)] = 4 \cdot 3/(s^2 + 9) = 12/(s^2 + 9).$$

Answer:
$$= \frac{12}{s^2 + 9}$$



30. Find the Laplace transform of $f(t) = 1 \cdot \sin(1 \cdot t)$.

$$\mathcal{L}\{1 \sin(1t)\}$$

→ Formula: $L[\sin(bt)] = b/(s^2 + b^2)$.

→ $L[1 \cdot \sin(1t)] = 1 \cdot 1/(s^2 + 1) = 1/(s^2 + 1)$.

Answer:
$$= \frac{1}{s^2 + 1}$$

